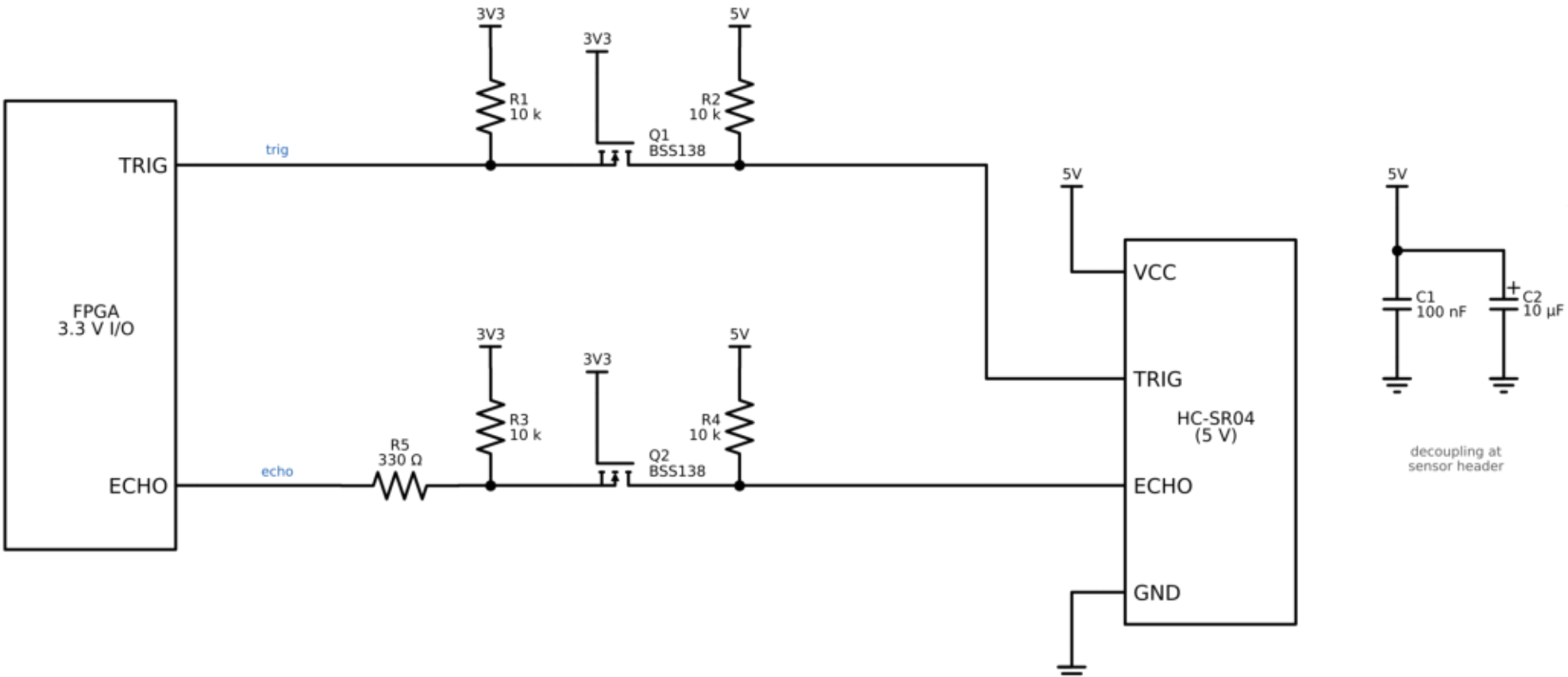
HC-SR04 interface - BSS138 bidirectional level shifter

Smart Radar Doorbell | Sheet 1 of 5

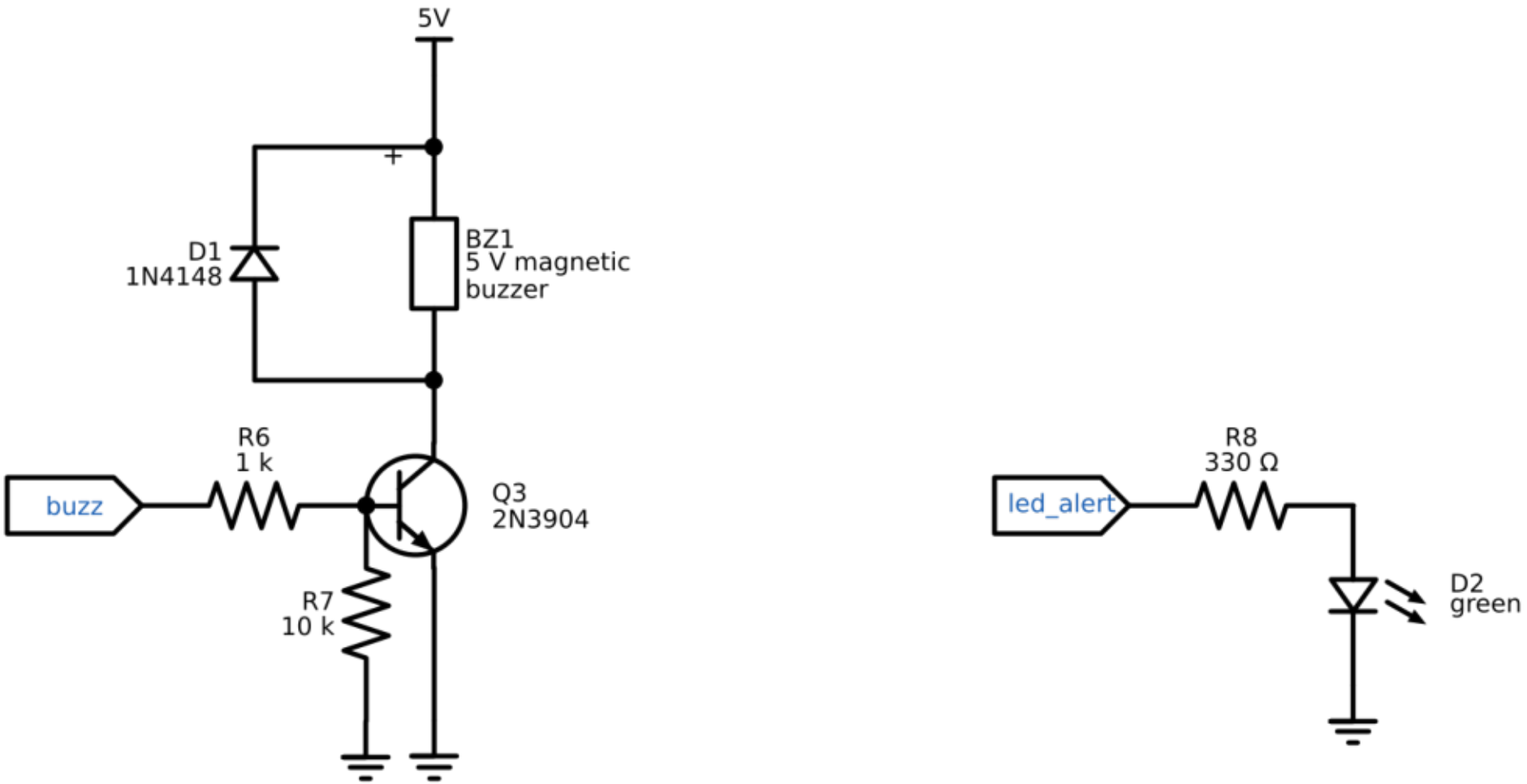
Gate tied to 3.3 V. Low side pulled to 3.3 V, high side to 5 V, so a 5 V ECHO never reaches an FPGA pin.



Buzzer driver and alert LED

Smart Radar Doorbell | Sheet 2 of 5

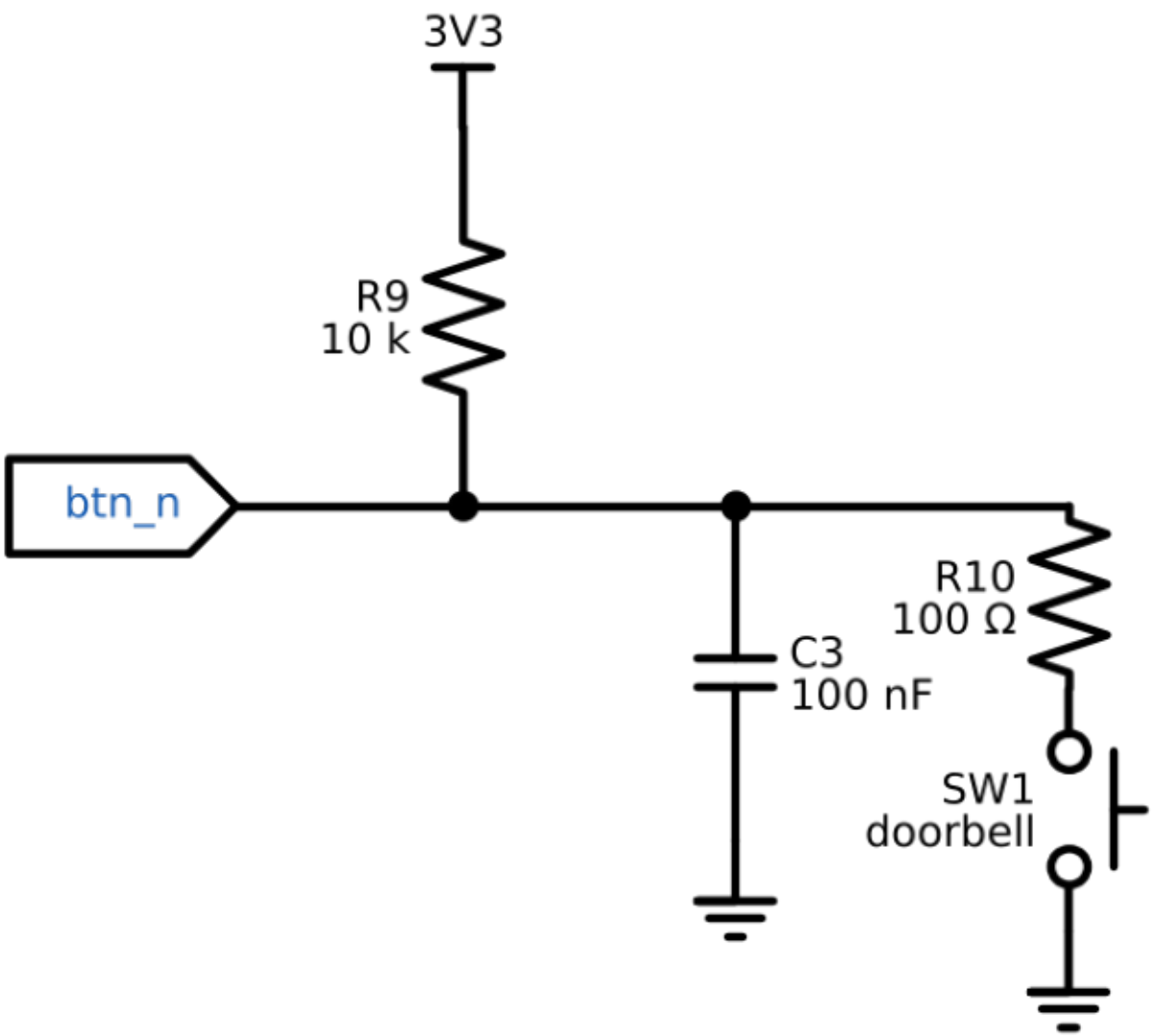
The FPGA cannot source 30 mA or switch 5 V, so an NPN low-side switch drives the buzzer.



Ib = (3.3 - 0.7) / 1 k = 2.6 mA; with a 30 mA buzzer that is a forced beta of about 12, so Q3 saturates hard.
D1 clamps the coil kick when Q3 turns off. R7 holds Q3 off while the FPGA is unconfigured.

Doorbell button input

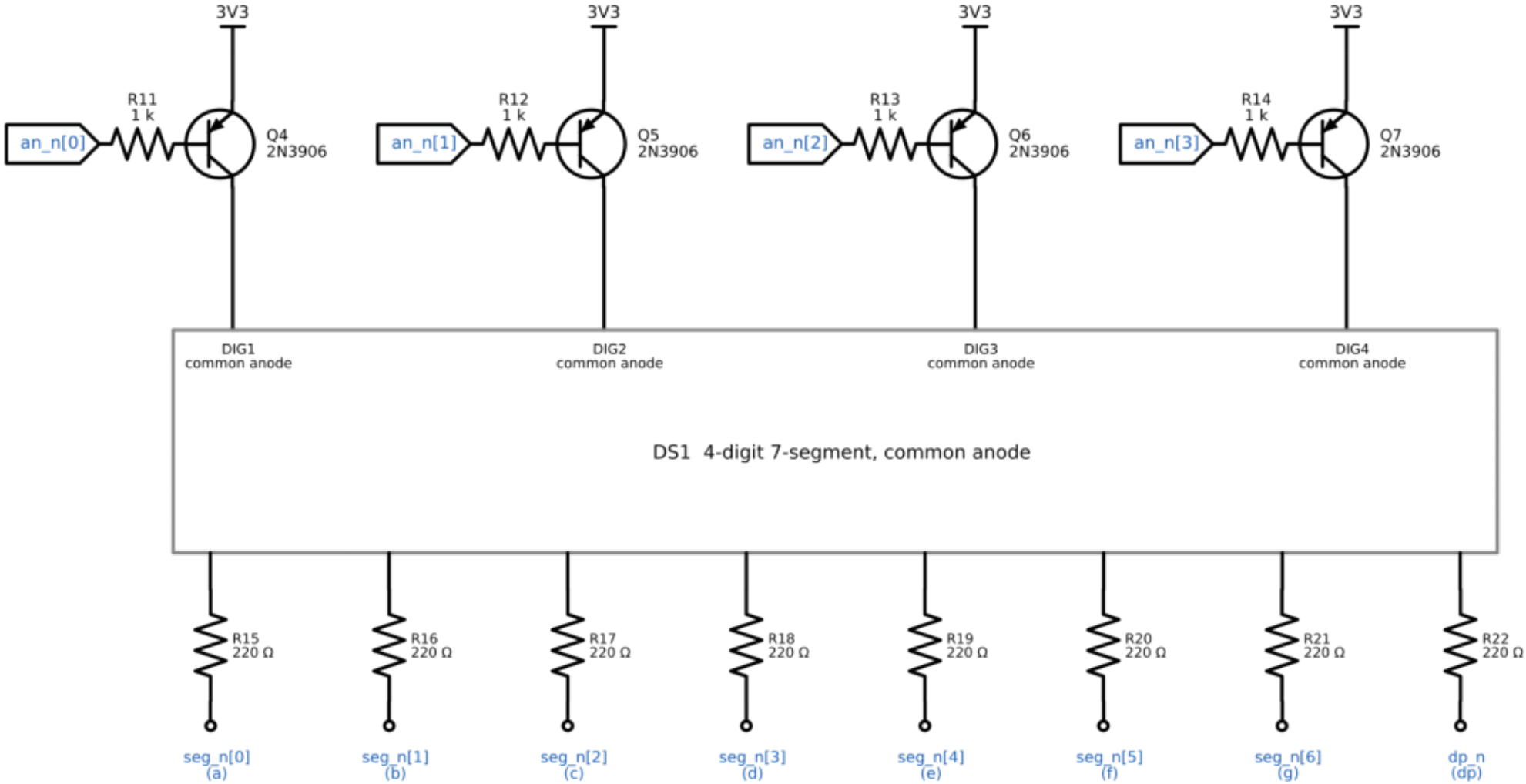
Smart Radar Doorbell | Sheet 3 of 5



R9·C3 = 1 ms knocks down fast contact chatter; R10 limits the cap discharge current through SW1. The real debounce is digital (rtl/debounce.sv): the input must hold for 10 ms before it is accepted.

Seven-segment display and PNP digit drivers

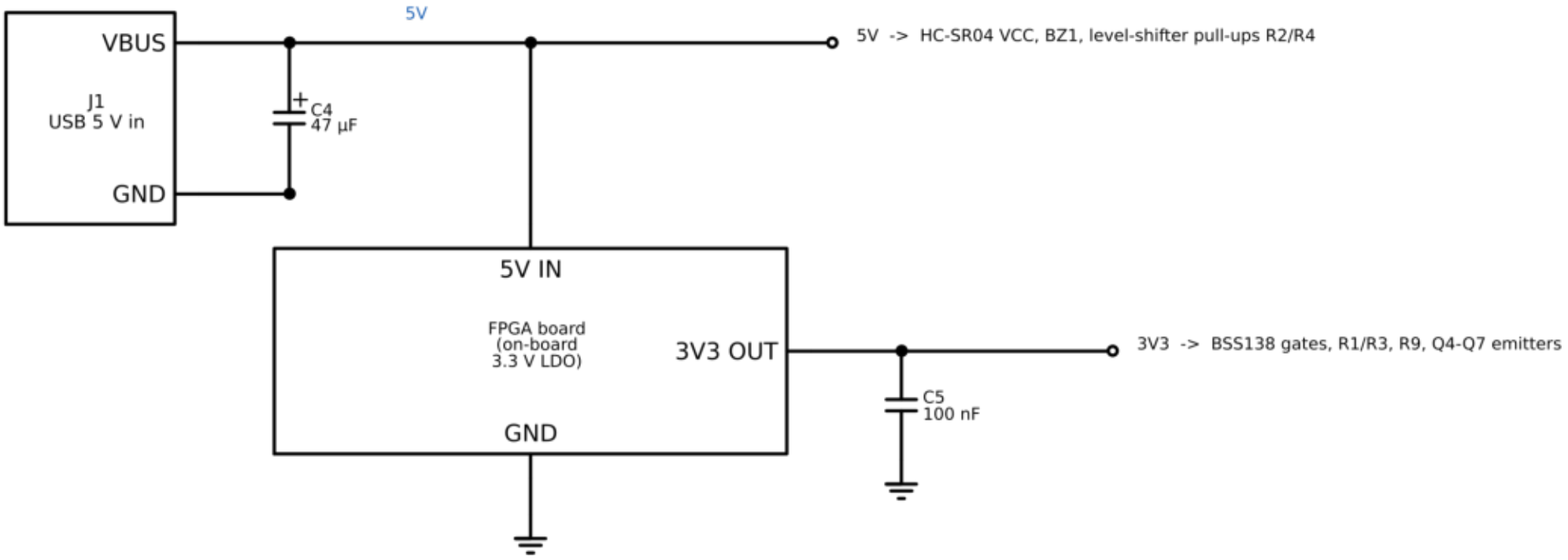
Smart Radar Doorbell | Sheet 4 of 5



Digits are scanned by rtl/sevenseg_mux.sv at 1 ms each (250 Hz frame). an_n low turns a PNP on; seg_n low sinks that segment. Segment current = (3.3 - 0.1 - 2.0) / 220 = 5.5 mA. The whole display runs from 3.3 V so no FPGA pin ever sees 5 V.

Power and grounding

Smart Radar Doorbell | Sheet 5 of 5



One ground, star-connected at J1. 3V3 only feeds logic-level parts; the only places 5 V and 3.3 V meet are the BSS138 level shifter (sheet 1) and the NPN collector (sheet 2), neither of which connects 5 V to an FPGA pin.